

NATIONAL INSTITUTE OF TECHNOLOGY PATNA

Department of Civil Engineering

Mid-Semester Examination, March-2023

Course Name: Hydrology and Irrigation Engineering

Course Code: CE43112

Exam Durations: 2 hours

B.Tech: 4th Semester

DOE: 04/03/2023

Full Marks: 30

Note: Answer all the questions. Mark for each questions are provided on the right side margin.

S.N.	Questions	Marks	CO	BL
1	Define Hydrology. Explain hydrological cycle mentioning all of its components followed by a neat sketch?	1+5=6	1	I,II
2	Assuming seepage loss to be 60% of evaporation loss, find the change in reservoir level for 10 hours in the next, if the water spread of the reservoir was 0.45 km ² . Given the following conditions- (a). Reservoir Catchment area = 10.5 km ² (b). One day observed uniform precipitation = 0.5 cm/h for 2h. (c). Runoff reaching the reservoir = 50%. (d). Outlet through canal = 1 m ³ /s (e). Rate of evaporation = 0.7 mm/hr/m ²	6x1=6	1	III
3	What do you mean by evapotranspiration? Explain in brief the various factors affecting evaporation?	2+4=6	2	I
4	Define probable maximum precipitation. A certain hydraulic structure has the life expectancy of 50 years and due to records of high precipitation it is designed for a flood magnitude of return periods of 50 years. What is the risk associated with this hydraulic design?	1+5=6	2	III
5	Write short notes on the following: (any three) a). Orographic precipitation. b). Transition-II type precipitation characteristics of India. c). Working mechanism of Weighing bucket type rain gauge. d). Depth-Area-Duration curve.	2x3=6	1,2	II

All the best